Sofiat Adeyemi

Phone – Email – Github – LinkedIn – Portfolio

EDUCATION

Southeastern Louisiana University (SLU) Hammond, LA
Bachelor of Science in Computer Science, Concentration: Data Science December 2026

- Thesis: Fine-Tuning LLMs using Reinforcement Learning

RELEVANT COURSEWORK

Core CS: Algorithms Design and Implementation, Software Engineering, Computer Architecture, Calculus, Linear Algebra, Statistics, Computing Systems
Data Science: Machine Learning, Database System, Data Mining, Artificial Intelligence, Computer Vision

RESEARCH EXPERIENCE

GPT-2 Model Training from Scratch – ***Stage 1 of Thesis - Deep Learning Research Project***

• Trained and optimized GPT-2-style language models from scratch using PyTorch and Hugging Face Transformers.
• Implemented tokenization, attention mechanisms, and gradient checkpointing for efficient model scaling.
• Followed deep learning best practices from Sebastian Raschka's *Deep Learning with PyTorch* series to improve convergence and model interpretability. We also used Sebastian Raschka's LLM from scratch on github.
• Conducted hyperparameter tuning, evaluation, and deployment using MLflow and AWS.

PROJECT EXPERIENCE

BookManagement - Online BookStore Database Hammond, LA

- Using SQL, we developed a database model for an online bookstore that allows users to browse, purchase, and review books.
- The database system will store book details and track purchase history. Book inventory will automatically update after sales, including when new stock is received.

Mood Classification with Machine Learning Hammond, LA

- Using Python and related libraries like Scikit Learn, we developed machine-learning models using ANN, SVM, DT, and KNN to predict the class of object labels.
- The process involved data preprocessing, feature extraction, model training, and evaluation to determine the most effective classification approach

TaskMaster – Time Management Web App Hammond, LA

- Served as a project team leader and developed a time management website.
- Incorporated C #, # ASP.NET, React, Typescript, Web API, Entity Framework, and semantic ui to create an interactive website

Student Wellbeing Data Analysis Hammond, LA

- Served as a project team leader, and we did an in-depth analysis of student issues.
- Incorporated Python libraries (NumPy, pandas, matplotlib, seaborn) to identify key issues students face in different districts and to understand student groups based on ethnic and social conditions.

Fraud Detection using Unsupervised Learning

- Built unsupervised **K-Means** and **Hierarchical Clustering** models to detect anomalous financial transactions.
- Cleaned and visualized data using **pandas**, **NumPy**, **matplotlib**, and **seaborn**.
- Identified unusual transaction patterns indicating potential fraud.

Traffic Sign Recognition Using A CNN made from scratch Hammond, LA

- Build a CNN architecture from scratch without using pre-trained models
- Achieve a classification accuracy above 93% on the test dataset
- Address real-world challenges like varying lighting conditions, weather effects, and partial obstructions
- Develop a strong model that can generalize to new traffic sign images

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TECHNICAL SKILLS

- **Programming Languages**: SQL, C, C#, Java, Python, TypeScript, Julia
- **Deep Learning:** PyTorch, TensorFlow, Keras, Hugging Face Transformers,
- **Machine Learning:** Scikit-learn, MLflow, OpenCV, Gradient Boosting, Feature Engineering
- **Data Science & Visualization:** Pandas, NumPy, MNE, Seaborn, Matplotlib
- **Web/Systems:** React, FastAPI, Flask, ASP.NET Core, REST APIs, Git, Multithreading, IoT, Docker
- **Cloud & Tools:** AWS, Google Colab, Kaggle, Jupyter Notebook, VS Code, Entity Framework.

- **Soft Skills:** Leadership, Communication, Technical Writing, Team Collaboration, Problem Solving, Adaptability

WORK EXPERIENCE

Computer Science Tutor Hammond, LA
SLU Computer Science Department August 2025 – Present

HONORS & INVOLVEMENT

- **Dean's List** - Fall 2023, **Honors List** - Spring 2024, **President's List** - Fall 2024, Spring 2025
- **ACM Member** - Fall 2023 - present